

1 §1.3 Direction Fields



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Lecture 3
§1.3 Direction Fields
Friday 11 September 2026

Announcements

- The public course web site is [Math 2C03: https://davidearn.github.io/math2c/](https://davidearn.github.io/math2c/)
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 lecture recordings and course announcements.
- Assignments and class polls are administered through [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, immediately after class.

Direction Fields

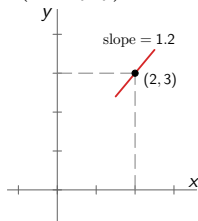
- Most nonlinear DEs cannot be solved exactly, *i.e.*, we cannot find an exact “quantitative” solution (we cannot find a formula for $y(x)$).
- But we can find approximate solution curves, using *qualitative methods*, without solving the problem exactly.
- A key qualitative tool is the *direction field* or *slope field*.

Direction Fields

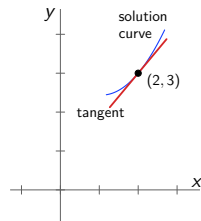
- **Recall:** If $f(x, y)$ and $\frac{\partial}{\partial y} f(x, y)$ are both continuous in a rectangle R , then there is a unique solution curve of $\frac{dy}{dx} = f(x, y)$ going through any point inside the rectangle R .
- At any point $(x, y(x))$ on a solution curve, the tangent line has slope $\frac{dy}{dx}$.
- We call a short segment of the tangent line through $(x, y(x))$ a **lineal element** at $(x, y(x))$.

e.g.,

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ &= 0.2xy\end{aligned}$$



(a) $f(2, 3) = 1.2$ is slope of lineal element at $(2, 3)$



(b) A solution curve passing through $(2, 3)$

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 2C03**
- Click on **Take Class Poll**
- Fill in poll **Slope of a lineal element**
- .

Direction Fields

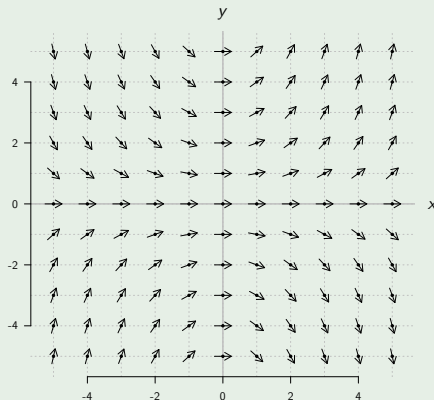
Definition (Direction Field)

A **direction field** of $\frac{dy}{dx} = f(x, y)$ is a collection of lineal elements plotted as arrows (pointing in the direction that the independent variable increases) on a grid of points in the plane.

- The direction field visually suggests the shape of a family of solution curves.

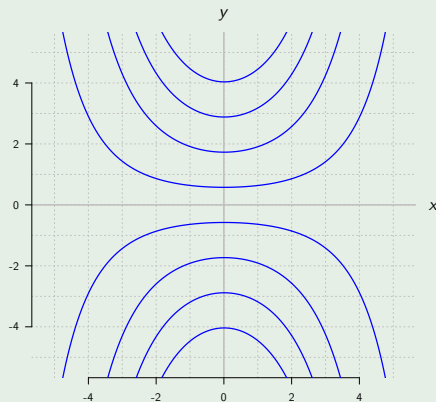
Direction Fields

Example (Direction field for $\frac{dy}{dx} = 0.2xy$)



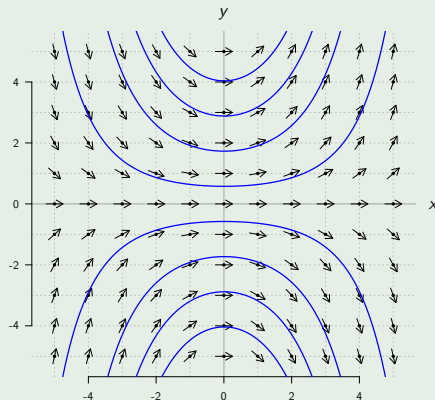
Direction Fields

Example (Solution curves for $\frac{dy}{dx} = 0.2xy$)



Direction Fields

Example (Direction field and solution curves for $\frac{dy}{dx} = 0.2xy$)



Direction Fields

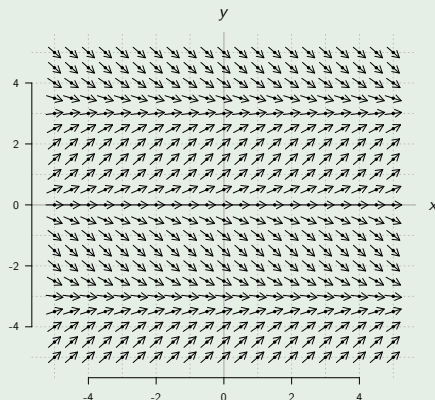
Now consider the first order ODE

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ &= \sin(y)\end{aligned}$$

- $f(x, y) = \sin(y)$ is continuous throughout the plane;
- $\frac{\partial}{\partial y} f(x, y) = \cos(y)$ is continuous throughout the plane;
- $\therefore$ there *is* a unique solution curve through every point in the plane.

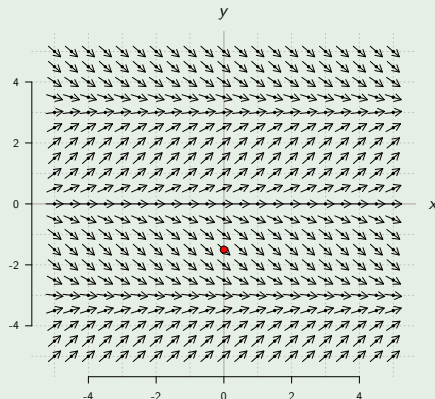
Direction Fields

Example (Direction field for $\frac{dy}{dx} = \sin(y)$)



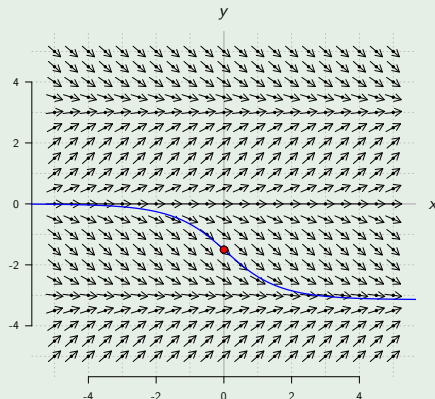
Direction Fields

Example (Direction field for IVP: $\frac{dy}{dx} = \sin(y)$, $y(0) = -\frac{3}{2}$)



Direction Fields

Example (Solution curve for IVP: $\frac{dy}{dx} = \sin(y)$, $y(0) = -\frac{3}{2}$)



Direction Fields

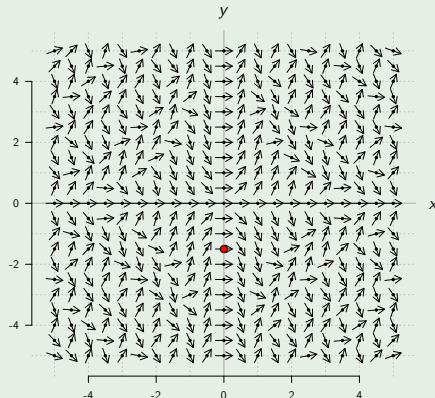
Now consider the first order ODE

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ &= 2\sqrt{1 + e^{-\sin(x^2+y^2)}} \sin(xy(1 + e^{-y^2}))\end{aligned}$$

- $f(x, y)$ is continuous throughout the plane;
- $\frac{\partial}{\partial y} f(x, y)$ is continuous throughout the plane;
- $\therefore$ there is a unique solution curve through every point in the plane.
- good luck finding exact solutions to this ODE...

Direction Fields

Example (IVP: $\frac{dy}{dx} = 2\sqrt{1 + e^{-\sin(x^2+y^2)}} \sin(xy(1 + e^{-y^2}))$, $y(0) = -\frac{3}{2}$)



Direction Fields

Example (IVP: $\frac{dy}{dx} = 2\sqrt{1 + e^{-\sin(x^2+y^2)}} \sin(xy(1 + e^{-y^2}))$, $y(0) = -\frac{3}{2}$)

